```
# 5. TRAINING THE MODEL
# ========================================
print(f"\nStarting training for {EPOCHS} epochs on GPU...")
history = model.fit(
    train_dataset,
    validation_data=test_dataset,
    epochs=EPOCHS
)
print("\nTraining Finished!")

# ========================================
# 6. ANALYZING PERFORMANCE
# ========================================
acc = history.history['accuracy']
val_acc = history.history['val_accuracy']
loss = history.history['loss']
val_loss = history.history['val_loss']
epochs_range = range(EPOCHS)

# Plot Training & Validation Metrics
plt.figure(figsize=(14, 5))

# Plot Accuracy Curve
plt.subplot(1, 2, 1)
plt.plot(epochs_range, acc, label='Training Accuracy', color='blue', lw=2)
plt.plot(epochs_range, val_acc, label='Validation Accuracy', color='orange', lw=2)
plt.title('Training and Validation Accuracy') .
plt.xlabel('Epochs')
plt.ylabel('Accuracy')
plt.legend(loc='lower right')
plt.grid(True, linestyle='--', alpha=0.6)

# Plot Loss Curve
plt.subplot(1, 2, 2)
plt.plot(epochs_range, loss, label='Training Loss', color='blue', lw=2)
plt.plot(epochs_range, val_loss, label='Validation Loss', color='orange', lw=2)
plt.title('Training and Validation Loss')
plt.xlabel('Epochs')
plt.ylabel('Loss')
plt.legend(loc='upper right')
plt.grid(True, linestyle='--', alpha=0.6)

plt.show()

# ========================================
```

Code

Modified

21m ago

Python 3 (ipykernel) | Idle

Mode: Command

Search

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Mostly sunny